



Toy Library case study

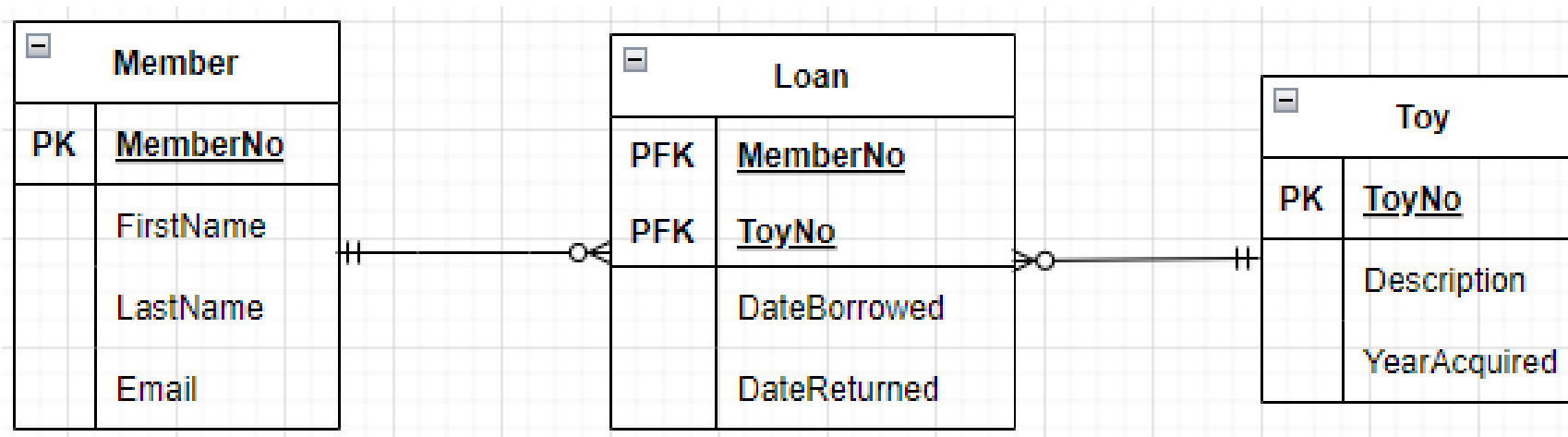
Additional explanation

Week 2 – Data modelling

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Conceptual Model



Member – Toy is M:M

Optional participation on both sides



Conversion to Relational Schema

The ERD translates to the following relational schema:

MEMBER (MemberNo, FirstName, LastName, Email)

PK: MemberNo

TOY(ToyNo, Description, YearAquired)

PK: ToyNo

LOAN(MemberNo, ToyNo, DateBorrowed, DateReturned)

PK: MemberNo, ToyNo

FK1: MemberNo references MEMBER

FK2: ToyNo references TOY

Is this correct??? Find out by testing with some sample data...



Testing the ERD

Generate some **sample data** for the **Strong** Entities

MemberNo 131	Louise Rossi	louise.r@gmail.com
MemberNo 74	James Chen	James_Chen@outlook.com

ToyNo 19	Tram with driver	2018
ToyNo 81	Tower crane	2016

Generate some **sample data** for the **Weak** Entity

<u>MemberNo</u>	<u>ToyNo</u>	DateBorrowed	DateReturned
131	19	1 June	8 June
131	81	1 June	8 June
131	19	19 July	30 July

Problem: duplicated PK



Testing Identifiers

The relational schema for Loan is now:

LOAN(MemberNo, ToyNo, DateBorrowed, DateReturned)

PK: MemberNo, ToyNo, DateBorrowed

FK1: MemberNo references MEMBER

FK2: ToyNo references TOY

Test the following Loan data rows. Will they be valid?

<u>MemberNo</u>	<u>ToyNo</u>	<u>DateBorrowed</u>	DateReturned
131	19	1 June	8 June
131	81	1 June	8 June
131	19	19 July	30 July
131	19	2 August	
74	19	11 June	18 June
74	81	11 June	18 June
74	19	31 July	30 August
131	81	9 October	



Keep testing...

Would the ERD still be correct if we included **DateReturned** as part of the Loan Identifier?

Three possible answers exist

- a. **DateReturned** should be part of Loan identifier
- b. **DateReturned** must not be part of Loan identifier
- c. **It makes no difference to the outcome / usage**

The relational schema for Loan is now:

LOAN(MemberNo, ToyNo, DateBorrowed, DateReturned)

PK: MemberNo, ToyNo, DateBorrowed, DateReturned

FK1: MemberNo references MEMBER

FK2: ToyNo references TOY

Test the Loan data rows. Will they be valid?



Testing Identifiers again

The relational schema for Loan is now:

LOAN(MemberNo, ToyNo, DateBorrowed, DateReturned)

PK: MemberNo, ToyNo, DateBorrowed, DateReturned

FK1: MemberNo references MEMBER

FK2: ToyNo references TOY

Test the following Loan data rows. Will they be valid?

<u>MemberNo</u>	<u>ToyNo</u>	<u>DateBorrowed</u>	<u>DateReturned</u>
131	81	1 June	8 June
131	19	19 July	30 July
131	19	2 August	

A PK must not contain Nulls

Row 3 fails this test.